

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that:

DNS requests are being sent from the computer to the DNS server using UDP on port 53.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message:

“UDP port 53 unreachable.”

The port noted in the error message is used for:

DNS services and DNS queries

The most likely issue is:

The DNS server's UDP port 53 is unreachable or no DNS service is listening on that port, preventing the computer from resolving the website's domain name to an IP address.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: The first incident occurred at **13:24:32 (1:24 PM)**. The same problem happened again at **13:26:32** and **13:28:32**.

Explain how the IT team became aware of the incident:

Several customers reported that they could not access the company website and received a **“destination port unreachable”** error. The analyst also tested the website and experienced the same problem.

Explain the actions taken by the IT department to investigate the incident:

The analyst used **tcpdump**, a network protocol analyzer, to capture and examine the network traffic when trying to access the website.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.):

The computer **192.51.100.15** sent DNS requests to the DNS server **203.0.113.2** using **UDP port 53**. The DNS requests were not delivered, and the DNS server returned an **ICMP “UDP port 53 unreachable”** error.

Note a likely cause of the incident: The **DNS service was unavailable or not listening on UDP port 53**, preventing the computer from getting the website's IP address. Therefore, the website could not be accessed.